

Report:

1. What LLM or similar product should we use to handle keyword generation?

We believe the Qwen-Instruct family is the best model suite for our use case, as it represents one of the most advanced lightweight model lines available. These models are specifically trained to handle technical inputs and consistently achieve higher benchmark scores compared to similar models in the same class. Among them, Qwen3-4B-Thinking-2507 would be the most ideal choice, as it is both the most up-to-date and the most powerful option. However, due to limited computational resources, specifically insufficient RAM and storage capacity, we are currently unable to test or deploy this model.

2. Try generating keywords for a consumer company, an industrials company, and a tech company. What are your thoughts on including all types of companies in this screener vs specific industries that we can better forecast? What would those industries/sectors be?

When generating keywords for different types of companies, such as consumer, industrials, and technology, it becomes clear that the usefulness of Google Trends data varies significantly by sector. Consumer-focused companies are the most suitable for this type of screening because their products and brands are directly searched by the public, making search interest a meaningful proxy for demand. Technology companies can be effective on a case-by-case basis. Firms that sell directly to consumers like hardware, consumer software, or subscription services, often generate substantial public search traffic, while companies focused on enterprise or government clients tend to have less representative search data. Industrial companies are generally less suitable for this approach, as their customers are typically other businesses or institutions rather than individual consumers, resulting in limited and less informative public search activity. Overall, a screener based on keyword generation and Google Trends will likely perform better if it focuses on specific, consumer-facing industries that are easier to forecast using public interest data. These industries could include consumer discretionary, consumer staples, retail, travel and leisure, and select areas of consumer technology. Limiting the scope to these sectors improves signal quality and reduces noise from industries where public search behavior is not a reliable indicator of underlying demand.

3. Play with prompt engineering for the keyword generation and give your top choice(s).

Updated Prompt: 10-K → High-Signal Google Trends Queries (Validated + Ranked, with Related-Demand Expansion)

Role

You are a financial analyst extracting demand-side signals from SEC filings to support revenue forecasting and screening design.

Source of Truth and Allowed Inference

Use the uploaded 10-K as the primary source of truth for what the company does, what it sells, and how it makes money. You are allowed to use general business intuition to infer the likely customer base and demand drivers based on the 10-K's product and business characteristics. You may use related terms that do not appear in the 10-K if they are logically connected to demand for the company's offerings.

Constraints (Critical)

1. Do not browse or use external sources.

2. Do not introduce products or markets that are unrelated to the company's described business model.

3. Every query must be tied to demand in one of two ways:

a) Direct: explicitly mentioned product, service, end market, use case, or workflow from the 10-K, or

b) Adjacent: complementary or substitute demand terms that would plausibly move with demand for the company's offerings (example: lawncare business can include "gardening supplies" as an adjacent demand proxy).

4. Avoid investor-only language (ticker, "EPS", "guidance", "earnings", "valuation", "price target", "stock").

5. Avoid HR, careers, and employee terms.

6. Prefer customer-searchable language over internal jargon.

Objective

Produce Google Trends queries that:

- Proxy for customer demand or customer evaluation
- Are plausibly linked to future revenue
- Help decide whether the company fits a broad cross-industry screener or should be limited to more predictable industries

What Counts as a High-Signal Query

A strong query reflects real external intent, such as:

- Searching for a product or service category the company sells or serves
- Comparing options, brands, providers, or alternatives
- Researching price, availability, replacement, repair, or adoption
- Responding to an operational or regulatory need that drives purchasing
- Capturing adjacent demand that plausibly moves with the company's demand even if the company is niche

Avoid:

- Generic macro terms unless clearly linked to a purchasing workflow
- Terms that could mostly represent non-customer interest
- Internal-only phrasing

Phase 0 (Internal Step): Build a Business Map from the 10-K

From the 10-K, infer and summarize internally (do not output) the following:

- Core offerings and revenue drivers
- Primary customer types and end markets
- Purchase workflow and channels (how customers find, evaluate, buy, service)
- Key complements and substitutes (adjacent demand terms)
- Any seasonal or cyclical demand patterns mentioned or implied by the business model

Phase 1: Generate Candidate Queries (Internal Step)

Generate up to 30 candidate Google Trends queries, split into two buckets:

- 15 to 20 Direct queries grounded in 10-K language and offerings
- 10 to 15 Adjacent queries based on complementary or substitute demand tied to the inferred business map

Query construction rules:

- Plain-language, realistic searches
- 2 to 6 words per query (only exceed if exact branded product name requires it)
- Mix of:
 - o Product or service category searches
 - o Problem-driven searches
 - o Workflow or industry searches if relevant
 - o Pricing, promo, availability, replacement, repair, maintenance, rental, or financing terms if applicable
 - o Competitor or alternative comparison terms
 - o Geography terms only if they matter to demand and are supported or implied by the business scope
- Must be searchable by external customers (not employees, not investors)

Phase 2: Validation, Scoring, and Selection (Internal Step)

For each candidate query, assign:

- query_type: direct or adjacent
- signal_strength score (1 to 5) based on clarity of customer intent and revenue linkage
- ambiguity_risk (low, med, high)

Then select the Top 10 strongest indicators based on:

- clarity of customer intent
- directness of revenue linkage
- specificity vs generality
- usefulness for deciding broad vs narrow screening
- balance between direct and adjacent queries (target 6 to 8 direct, 2 to 4 adjacent, but adjust if the company is very niche)

Final Output (STRICT JSON ONLY, no prose, no markdown)

Return STRICT JSON ONLY. Output exactly 10 objects with fields:

```
{ "query": "", "segment": "", "revenue_link": "", "geo": "", "ambiguity_risk": "low|med|high", "query_type": "direct|adjacent" }
```

Field requirements:

- query: exact Google Trends query (2 to 6 words)
- segment: inferred customer or end-market segment based on the 10-K product and business characteristics (use 10-K wording when possible)
- revenue_link: one concise sentence explaining how this query connects to revenue, referencing either direct demand or adjacent demand logic
- geo: geographic scope; if not stated or not relevant, use "unspecified"
- ambiguity_risk: low, med, or high
- query_type: direct or adjacent

Failure Condition

If the company is extremely niche and you cannot create enough direct queries, increase the share of adjacent queries. Do not fabricate unrelated markets. Output exactly 10 final objects even if you have to rely more heavily on adjacent demand proxies.

Reason: Initially, we used both Gemini and ChatGPT to generate prompts and tested them across the same tasks. We found that ChatGPT's prompts were more detailed and consistently led to higher-quality model outputs. In particular, ChatGPT generated prompts that restricted outputs to words and phrases explicitly found in the companies' 10-K reports, rather than inferring or introducing terminology based on assumed product offerings or business characteristics.

4. Is there another financial statement or document we can easily access that you think would give good additional context for keyword generation?

10-Q reports: Adding the most recent 10-Q, especially the MD&A and segment discussion, gives better context for keyword generation because it is usually more current than the annual report. It also explicitly describes what is driving or hurting revenue right now. It often breaks out volume versus price, backlog of orders, customer demand trends, channel behavior, and regional or segment shifts. That language tends to be closer to how customers and buyers describe the business, which helps produce search queries that are more timely and demand linked.

8-K Reports: An 8-K earnings release (Item 2.02 and its exhibit) is also a strong supplement because management typically summarizes performance drivers in

plain, high-signal terms. It often highlights which products or categories saw stronger demand, where pricing actions occurred, which end markets were strong or weak, and what operational constraints mattered. Since it is short and focused on the quarter's main themes, it can surface customer-facing phrases and product terms that are more likely to show up in Google Trends without wading through broader disclosure.

5. Any other thoughts on the investment idea or process.

For the overall investment idea, the primary driver of success will be minimizing noise while maintaining a repeatable workflow. We also believe that, to maximize the likelihood of success, access to a premium version of Google Colab would be necessary, as it would enable the use of more advanced models that require greater storage capacity and computational resources.